

In this problem set, the questions marked with a (★) are required for students enrolled in 583 and bonus for students enrolled in 483.

1. A three-dimensional spacetime has the line element

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\phi^2. \quad (1)$$

- (a) (10 points) Find the explicit Lagrangian for the variational principle for geodesics in this spacetime in these coordinates.
- (b) (10 points) Using the results of (a) write out the components of the geodesic equation by computing them from the Lagrangian.
- (c) (5 points) Read off the nonzero Christoffel symbols for this metric from your results in (b).

2. **Christoffel Symbols, aka connection coefficients** Consider a change of basis $\tilde{e}_\mu = A^\nu_\mu e_\nu$.

- (a) (10 points) Show that the components of the Christoffel symbol in the new basis are related to the components in the old basis by

$$\tilde{\Gamma}^\mu_{\rho\nu} = A^\sigma_\rho (A^{-1})^\mu_\tau [A^\lambda_\nu \Gamma^\tau_{\sigma\lambda} + \partial_\sigma A^\tau_\nu]. \quad (2)$$

Given the above equation, are the Christoffel symbols tensors? Why or why not?

- (b) (10 points) Does the difference of two connections, $S^\mu_{\rho\nu} = (\Gamma_1)^\mu_{\rho\nu} - (\Gamma_2)^\mu_{\rho\nu}$ obey the tensor transformation law?
3. (15 points) (★) An observer in a circular orbit of circumference $2\pi R$ around a charged, spherical, black hole (Reissner-Nordström black hole) of mass M and charge q measures local electric and magnetic fields. What are their strengths and orientations? The Reissner-Nordström metric is given by

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad (7)$$

where $f(r) \equiv 1 - 2GM/r + q^2/r$.